

Pain in the Achilles region

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The anger is rage. Why the hell did this happen!!? . . . Now I'm supposed to come back from this and be the same player or better . . . How in the world am I supposed to do that?? . . . Maybe this is how my book ends. Maybe Father Time has defeated me . . . Then again maybe not!

Kobe Bryant, LA Lakers shooting guard and five-time NBA champion, after rupturing his Achilles tendon in a game against the Golden State Warriors in 2013. He returned to play until his retirement in 2016

The Achilles tendon is prone to injury when subjected to unaccustomed repetitive high loads.^{1,2} Runners, for example, have a 15 times greater risk of Achilles tendon rupture and 30 times greater risk of tendinopathy as they age than do sedentary controls.³ In this chapter, we review the approach to assessment (history, examination, investigation). We then outline the specific treatment approaches.

PRACTICE PEARL

Patients most commonly present with gradual (overuse) pain in the Achilles region due to pathology in the midportion of the Achilles tendon (Fig. 40.1a).

Other important conditions in order of prevalence at a sports clinic with a broad-based clientele are:

- pain at the insertion of the Achilles tendon and associated retrocalcaneal bursa ('enthesis organ')
- posterior impingement (an important differential diagnosis)
- Achilles pain related to the plantaris tendon
- pain related to friction between the Achilles tendon and surrounding tissues (peri-tendinopathy).

Achilles tendon pathology can also manifest in a very different, dramatic manner: via acute rupture of the Achilles tendon. This immediately disabling condition is captured in the chapter opening quote. The management of cases like that of Kobe Bryant is discussed in the last section of this chapter.

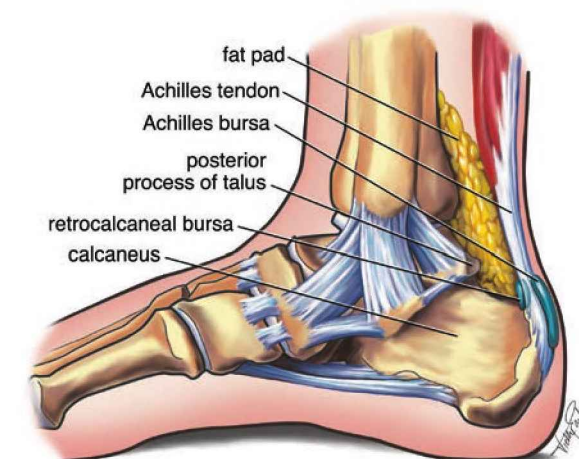
CLINICAL PERSPECTIVE

The key areas of pain in the Achilles region (posterior heel and proximal toward the calf) are illustrated in Figure 40.1.

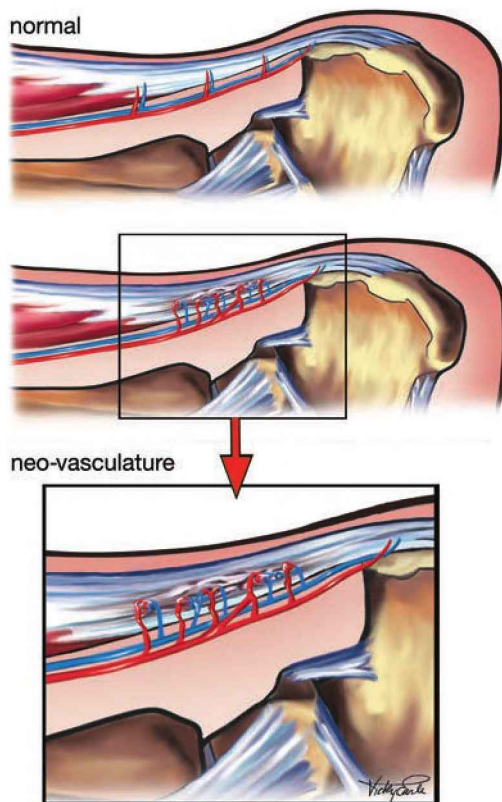


(a)

Figure 40.1 The Achilles region (a) The midportion of the tendon is clearly distinct from the insertion—an important clinical feature



(b)



(c)

Figure 40.1 (cont.) (b) The anatomy of the Achilles midportion, Achilles insertion and posterior process of talus (c) Stylised illustration to show the blood supply of the normal Achilles tendon (top) and in tendinopathy (middle and below)

The Achilles tendon is the combined tendon of the gastrocnemius and soleus muscles, and is the thickest and strongest tendon in the human body.⁴ Anteriorly, the deep surface of the tendon is supported by a fat pad (Fig 40.1b) through which most of the vessels and nerves enter the tendon (Fig. 40.1c). The tendon has no synovial sheath, but has a peritendon which is continuous with the perimysium of the muscle and the periosteum of the calcaneus, and which enfolds the anterior fat pad of the tendon. The tendon attaches to the inferior half of the calcaneus. The plantaris tendon is located medial to the Achilles and has variable anatomy, which is discussed later. The posteriosuperior process of the calcaneus approximates the Achilles, usually with a bursa between.

The vast majority of patients who present with gradual onset of Achilles region pain can be treated successfully non-operatively. Experienced clinicians can quickly home in towards a diagnosis by following the clinical reasoning outlined in Figure 40.2. The first step is to rule out recent acute rupture by listening to the patient's history and performing one simple clinical test (see Physical examination, below). In many cases of acute Achilles tendon rupture, the patient or a friend has diagnosed the condition before reaching the clinician's office.

Among the conditions that manifest with more gradual onset of Achilles region pain, the most common is midportion tendinopathy. Have the patient point to the location of pain during the history—that part of the leg is easily accessible during the office consult with most types of clothing. If the history and location of pain do not fit the typical pattern for midportion tendinopathy, use the location of pain to consider other diagnoses. Broadly speaking, the second most prevalent condition is insertional tendinopathy and its associated bursitis, but in settings like a ballet school, posterior impingement will be more common than insertional tendinopathy (and more common than midportion tendinopathy too!).

History

The standard presentation is with the patient reporting a gradual onset of pain around the midportion of the Achilles, often first noticed on waking and generally associated with a recent increase in load volume or intensity.¹ Patients complain of pain at the start of activity and pain and stiffness the following morning.

PRACTICE PEARL

The combination of morning stiffness and pain confined to the tendon is a hallmark of Achilles tendinopathy. If these are not present, the clinician should consider alternative diagnoses.

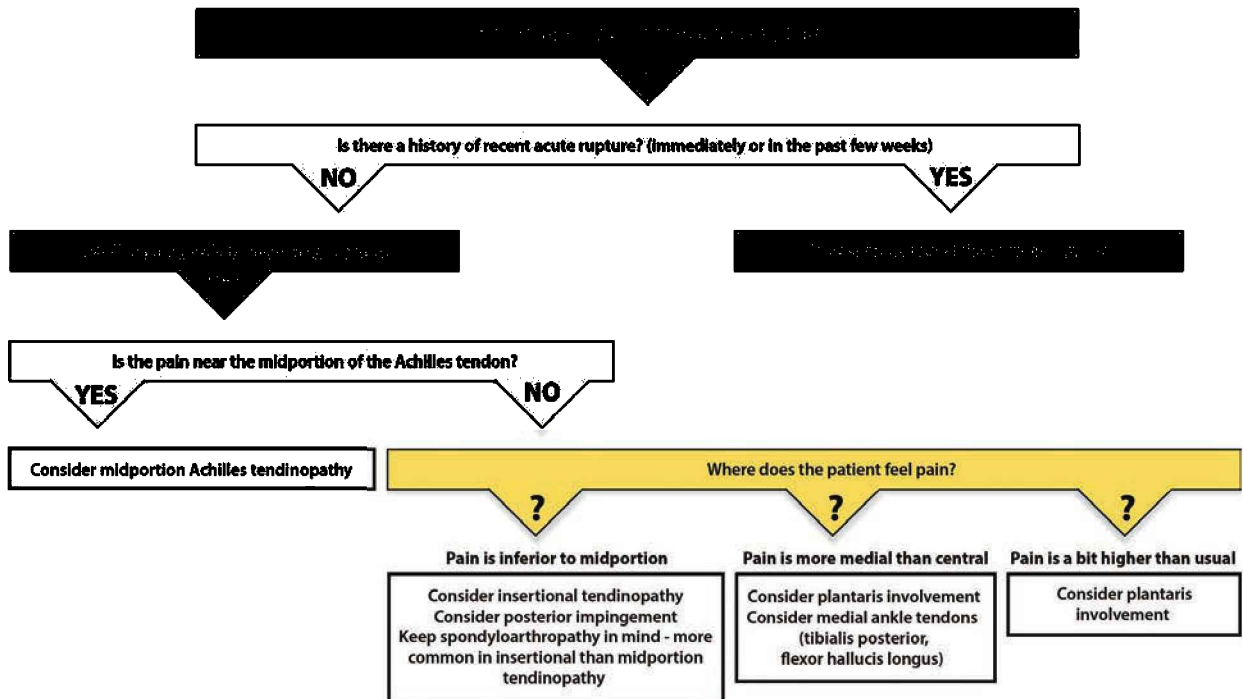


Figure 40.2 Flow diagram showing how to rapidly subgroup patients with various presentations of pain in the Achilles region

If the patient gives a history of typical midportion Achilles pain, the clinician can follow up with questions aiming to identify the cause. Was the main inciting factor a sudden increase in loading—for example, due to an increased volume, intensity or type of exercise, or a change in surface or footwear? Commonly, return to loading after a layoff for another injury is the catalyst; this also reflects a relative increase in load. Or is there a systemic condition underpinning tendinopathy (e.g. consider the possibility of Type 2 diabetes mellitus, especially in older and overweight people)?

Careful history taking is important in case the gradual-onset Achilles region pain arises from neighbouring anatomy. For example, pain medial to the Achilles tendon or deep to the Achilles tendon with a pain pattern that is not typical of Achilles tendon suggests an alternative diagnosis.

As highlighted in Figure 40.2, the location of pain provides important diagnostic information. Pain closer to the musculotendinous junction than the standard midportion Achilles pain (Fig. 40.1a) can arise from the plantaris tendon being pathological itself or causing local Achilles tendon pathology.

Table 40.1 Causes of pain in the Achilles region

Common	Less common	Not to be missed
Midportion Achilles tendinopathy (including tendinosis—with or without partial tears—and peritendinopathy)	Plantaris tendon involvement	Achilles rupture—partial/total accessory soleus muscle
Insertional Achilles tendinopathy (at the superior lateral calcaneum, pathology includes retrocalcaneal bursitis)	Flexor hallucis longus and tibialis posterior tendinopathy (Chapter 42)	Referred pain • Peripheral neural structures • Lumbar spine
	Subcutaneous bursitis	Sever's disease (in adolescents)
	Posterior ankle joint impingement	Metabolic condition (potentially undiagnosed) • Hypercholesterolaemia • Type 2 diabetes mellitus
	Tightness of the posterior Achilles retinaculum, compressing the tendon (rare)	Spondyloarthropathy (Chapter 7)

PRACTICE PEARL

If the patient offers a history of repeated soleus strains (low calf pain that comes and goes), has proximal, *medial* Achilles pain and is intolerant of dorsiflexion loading, be alert to the possibility of plantaris tendon involvement.

If the patient engages in activities that involve repeated end-range plantarflexion, posterior impingement is the key differential diagnosis for pain approximately at the midportion. The textbook activity with high prevalence of this condition is ballet (pointe/demi-pointe), but it is a common presentation to the clinic in football players (kicking), sprinters and cricket fast bowlers.

A key alternate diagnosis

Anatomically, the impingement (of the os trigonum or the posterior process of the talus, Fig 40.10 below) takes place anterior and inferior to the Achilles midportion. Some patients appreciate that the pain feels 'deeper' than at the Achilles tendon, but many do not offer that distinction, so the clinician must bracket the midportion Achilles diagnosis with the possibility of posterior impingement. It is not uncommon for a patient to have wasted time and 'failed' prolonged Achilles rehabilitation when the true diagnosis was posterior impingement. Having considered these two conditions together, the clinician can easily distinguish them during physical examination (see passive plantarflexion).

The patient may report pain medially at about the level of the Achilles tendon; this can be due to tibialis posterior peritendinopathy (proximal to the flexor retinaculum of course) or flexor hallucis tendinopathy (a common ballet pathology). These two conditions generally present more inferiorly and more medially than typical midportion Achilles tendinopathy. These two conditions are discussed in Chapter 42.

If the patient gives a history of relatively unloaded plantarflexion and dorsiflexion, such as an inexperienced cyclist spiking his or her load, consider peritendinopathy.

A sudden explosive event

A sudden, disabling sensation or pain and inability to function is often due to complete tendon rupture. The greatest prevalence is among men aged between 30 and 50.^{5,6} It occurs in sports with explosive movements like basketball, tennis and Ultimate (Frisbee®), but it can arise in people with underlying degenerative (but asymptomatic) tendon pathology as simply as a result of stepping off a curb awkwardly. The patient often reports hearing a noise like a gunshot or feeling like he or she was kicked on the tendon.

PRACTICE PEARL

Many textbooks suggest that complete Achilles rupture prevents active plantarflexion. This is wrong! Patients can often plantarflex weakly in non weightbearing positions using intact accessory plantar flexors such as tibialis posterior, plantaris, flexor digitorum longus and flexor hallucis longus. It is a naive error to rule out Achilles tendon rupture from patient reports of plantarflexion. The key diagnostic test is the 'calf squeeze test' (also called Simmond's or Thompson's test)^{7,8} (see Fig. 40.3h).

When the patient reports an acute episode, other possibilities include (i) partial rupture of the Achilles tendon and (ii) plantaris tendon rupture. The latter presents with acute onset of pain on the medial side of the upper Achilles midportion. In athletes, this typically occurs during sprinting. If the presentation to the clinic is not acute, the patient may report having had a short period (3–5 days) with disabling pain and then being able to run again.

Physical examination

If there is no suspicion of tendon rupture, the examination should begin with the patient standing. The clinician should first assess the muscles for signs of wasting, then examine how the patient copes with loading tasks likely to provoke pain. In most patients, simple double or single-leg heel raises will cause pain. In more active individuals whose tendon can tolerate higher loads, ask the patient to hop on the spot or hop forward to reproduce pain.

In some athletes, repeated loading tests (multiple hops, jumps) or examination after a training session will be valuable to fully evaluate the tendon. Record not only the function of the calf and Achilles tendon, but also note how the lower limb kinetic chain is performing. Longstanding symptoms can cause substantial unloading (favouring the unaffected side) and impair the function of the entire kinetic chain (see Chapter 8).

After assessing function, assess strength and strength endurance, and seek possible predisposing factors such as unilateral calf tightness, reduced dorsiflexion and abnormal lower limb biomechanics. This may require analysis of treadmill running or video analysis of jump and other sporting movements that are provocative (Chapter 8).

Finally, compare the resting muscle tone between sides and palpate around the tendon for specific areas of tenderness (e.g. ventral, superficial, medial tendon), crepitus (a 'crackling' feeling that arises because of the fibrinous exudate in the peritendon) and tendon thickening.

PRACTICE PEARL

Palpation tenderness in the Achilles is not a diagnostic or prognostic sign. Achilles tendons are always sore to touch. (Tenderness—pain on palpation—has a very poor positive likelihood ratio (LR+). See Chapter 14.)

B

A focused examination relevant to the patient who presents with Achilles pain includes:

1. Observation

- standing and walking: Observe soleus and gastrocnemius muscle bulk on both sides. Look for wasting of quadriceps femoris and gluteal muscles
- ask the patient to identify the location of pain (midportion, distal, specific to Achilles tendon or more medial/lateral) (Fig. 40.1a). You may see thickening at site of pain or you may be able to feel crepitus
- if Achilles tendon rupture is a possibility, inspect for obvious swelling (Fig. 40.3a). If the patient is seen acutely, the tendon defect is evident beneath the skin at the site of rupture. This defect fills rapidly with blood but if the patient is lying supine, the unsupported foot rests in excessive dorsiflexion due to loss of resting calf muscle (plantarflexion) tone (see also special tests, below).

2. Functional tests

- plantarflexion—double- and single-leg calf raises for pain
- hopping/forward hopping (if clinically indicated—in higher-functioning patients while testing their capacity) (Fig. 40.3b)
- strength/endurance—examine the patient's capacity to repeat single-leg heel raises with full height. Record the number as a baseline
- biomechanics (see also Chapter 8 for general biomechanical assessment)
 - Active dorsiflexion (lunge)
 - Gastrocnemius stretch (Fig. 40.3c)
 - Soleus stretch (Fig. 40.3d)

3. Supine/prone

- active movements (only useful if rupture suspected)
 - plantarflexion and dorsiflexion
 - first-toe flexion and extension in full ankle plantarflexion for flexor hallucis longus tendinopathy (Fig. 40.3e)



(a)



(b)

Figure 40.3 Examination of the patient with pain in the Achilles region (a) Inspect for Achilles tendon rupture if this is a possibility. In this case the right side has a ruptured Achilles tendon (b) Functional tests. These can be used to reproduce pain, if necessary, or to test strength. Tests include double-leg and single-leg calf raises, hops (illustrated), hops forward, eccentric drops and lunge



(c)



(d)

Figure 40.3 (cont.) (c) Passive movement—muscle stretch (gastrocnemius). The patient stands so that body weight causes overpressure. The knee must remain extended and the heel (back foot) keeps contact with the floor. The foot remains in neutral if the patient keeps the patella in line with the second metatarsal. Compare the stretch on both sides (d) Passive movement—muscle stretch (soleus). The patient stands upright and keeps the knee flexed. The foot should remain in a neutral position

b. Passive movements

- i. plantarflexion with and without overpressure (Fig. 40.3f) to assess posterior impingement)
- ii. inversion and eversion

c. Palpation

- i. firm palpation of the Achilles tendon is not recommended as normal tendons are tender—they are painful to touch
- ii. palpation may identify associated pathologies (such as the more proximal medial tenderness

if the plantaris is a cause of pain/contributing to Achilles tendinopathy)

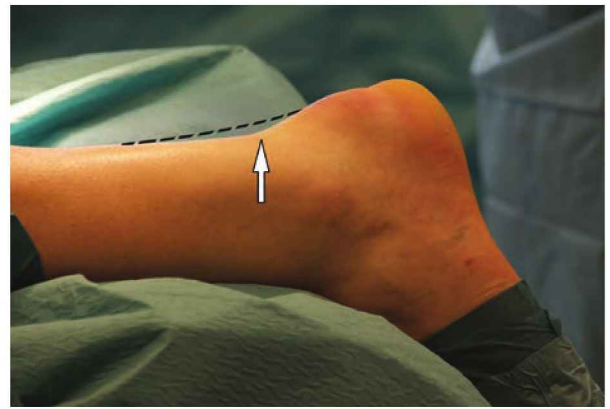
- iii. Posterior talus may be tender in posterior impingement but the key test is the passive plantarflexion test
- iv. Palpate more inferiorly and medial to the midportion Achilles tendon for flexor hallucis longus tendinopathy
- v. Palpate the tibialis posterior tendon before it passes under the flexor retinaculum to ensure it is not contributing to symptoms
- vi. Assess the flexor hallucis longus, which is a differential for posterior impingement



(e-i)



(e-ii)



(g)



(f)

Figure 40.3 (cont.) (e) Active first-toe flexion and extension in full ankle plantarflexion for flexor hallucis longus tendinopathy (e-i) Active movement without resistance (e-ii) Movement against the examiner's resistance. Examiner is feeling for FHL swelling or crepitus with her left hand (f) Passive movement—plantarflexion. This will be painful if posterior impingement is present. Overpressure can be applied (g) Observation (prone)—for suspected Achilles tendon rupture. The Achilles tendon would normally bridge the space from the superior calcaneum through to the muscle tendon junction (normal anatomical skin/tendon position shown by dotted line). In cases of tendon rupture (arrow), the tendon is not connected to the calcaneum and rests on underlying soft tissue. (h) Special test—the calf squeeze test. (h-i) At rest. The ruptured Achilles (right leg, white arrow) hangs more vertically compared to the uninjured leg (red arrows). (h-ii) Squeeze. The ruptured Achilles (right leg) fails to plantar flex (red arrow), while the normal leg plantar flexes on squeeze (green arrow)



(h-i)



(h-ii)

- vii. Achilles tendon insertion and retrocalcaneal bursa (insertional tendinopathy)
- viii. Tightness in the posterior retinaculum
- ix. Crepitus associated with peritendinopathy

4. Special tests

- a. Inspect for tendon defect and loss of resting tone for tendon rupture (Fig. 40.3a, g)
- b. Simmond's/Thompson's calf squeeze test^{7, 8} (Fig. 40.3h).

Key outcome measures (PROMs)

Functional tests mentioned and depicted above (number of calf raises, hops) provide a baseline against which to compare treatment response. In addition, the Victorian Institute of Sport Assessment–Achilles (VISA-A) has been widely used since 2001 to monitor the clinical progress of Achilles tendinopathy¹⁷ (Table 40.2). It takes less than 5 minutes to complete and once patients are familiar with it they can complete it themselves. Scores range from 0–100, where 100 represents no pain, full function and no limitation of training or competition. Baseline values for Achilles tendinopathy patients can be expected to be between 24 and 60, and scores above 90 represent full recovery.¹⁸ The minimum clinically important difference (MCID) of the VISA-A score is in the range of 12–20.^{19–21} The VISA-A should be administered monthly rather than daily or weekly as 40% of the score comes from capacity to complete higher-load activities.

Investigations

Plain radiographs are of limited value—they do not improve the positive likelihood ratio (Chapter 14) of making the diagnosis of a form of Achilles tendinopathy (either midportion or insertional).²² Tendon calcification is not a predictor of current symptoms or prognosis (Fig. 40.4a). The patient should be managed according to symptoms. A radiographic finding of a prominent superior projection of the calcaneus (Haglund's morphology) is normal. See discussion of this condition below. Anatomical changes responsible for posterior impingement may also be evident on plain radiographs (see posterior impingement section later in this chapter); such anatomy does not predict current symptoms or prognosis.

In patients with Achilles tendon pain, ultrasound examination can help distinguish whether the Achilles or the peritendon or both have pathology. Greyscale ultrasound examination (Fig. 40.4b, c) can be combined with colour and power Doppler (Fig. 40.4d) that reflects blood flow inside and outside the tendon. MRI can demonstrate abnormal tendon morphology (Fig. 40.4e) but it did not prove a good predictor of current or future symptoms.¹⁵

PRACTICE PEARL

Under careful diagnostic scrutiny, imaging such as colour and power Doppler examinations and MRI have been disappointing for clinicians who had hoped they could offer prognostic information or guide response to treatment^{15, 23–25} (see also Table 40.4 and Chapter 15).

Ultrasound tissue characterisation (UTC) (Fig. 40.4f) is not grouped with ultrasound modalities above as it is not in wide clinical use. It is an extension of traditional ultrasound¹⁹ (Chapter 15) that allows intratendinous structure to be quantified. As with other imaging modalities, an abnormal UTC appearance does not guarantee that the tendon is the source of pain; experienced clinicians believe that the appearance of abnormal collagen may increase the positive likelihood ratio (Table 40.3). UTC can demonstrate changes in Achilles tendon structure with loading rehabilitation (i.e. the effect of mechanotherapy, Chapter 17),²³ in cases of tendinopathy and also after tendon rupture. UTC is not yet useful to guide treatment; rehabilitation should be guided by clinical assessment and functional testing.

MIDPORTION ACHILLES TENDINOPATHY

This section outlines (i) pathology so that it can be explained to patients, (ii) relevant risk factors and (iii) treatment principles including a four-stage exercise program based on the concept of tendon capacity (Chapter 4, Fig. 4.14).

Pathology

The macroscopic changes seen at surgery for chronic midportion Achilles tendinopathy are characterised by a poorly demarcated dull-greyish discolouration of the intratendinous collagen with a focal loss of normal fibre structure.^{26, 27} Partial ruptures may occur in regions of pre-existing pathology.²⁸ Peritendinous structures may be normal, oedematous or scarred. If scarred, the peritendon may adhere to the tendon.

Under the light microscope, pathological tendon reveals abnormal proliferation of tendon cells (fibroblasts), increases in ground substance, collagen fibre disarray and increased vascularity as outlined in Chapter 4 (see box and Figs. 4.10 and 4.11). These regions of tendon disarray correspond to areas of increased signal on MRI and hypoechoic regions on ultrasound²⁹ (Fig. 40.4c, d, e).

B

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Table 40.3 Diagnostic accuracy of clinical tests in patients with tendinopathy and rupture

Assessment finding	What is a positive finding?	Sensitivity	Specificity	Likelihood ratio –	Likelihood ratio +
History					
Tendinopathy					
Pain ⁹	Patient points to site	78	77	0.50	3.39
Morning stiffness ⁹	Pain worse in morning	89	58	0.19	2.12
Physical examination					
Tendon rupture					
Palpation of gap in tendon ¹⁰	Tendon gap	73	89	0.30	6.64
Tendinopathy					
Palpation (tenderness) ¹¹	Patient reports tenderness	64	81	0.48	3.15
Palpation of tendon thickening ⁹	Subjective tendon thickening	59	90	0.46	5.9
Palpation of crepitus ⁹	Subjective crepitation with passive ankle movement	3	100	0.97	>10*
Arc sign ¹¹	Swelling moves during ankle dorsi/plantarflexion	42	88	0.68	3.24
Special tests					
Tendon rupture					
Thompson's calf squeeze test ⁷	Ankle remains still	40	–	–	–
Calf squeeze test ¹⁰	Ankle remains still	96	93	0.04	13.71
Matles test ¹⁰	Neutral or dorsiflexed ankle	88	85	0.14	6.29
Tendinopathy					
Royal London Hospital test ¹¹	Tenderness on palpation in end-range dorsiflexion	54	86	0.54	3.84
Imaging					
Tendon total rupture					
Ultrasound ¹² Radiologist interpretation		100	83	0	5.88
MRI ¹³ Radiologist interpretation		91	–	–	0.91
Tendon partial rupture					
Ultrasound ¹⁴ Radiologist interpretation		50	81	0.62	2.63
Tendinopathy					
Ultrasound ¹⁵ Radiologist interpretation		80	49	0.39	1.57
MRI ¹⁵ Radiologist interpretation		95	50	0.1	1.9
Combination of findings					
Tendinopathy¹⁶					
+ Palpation (tenderness)	3/3	59	83	0.49	3.47
+ Arc sign					
+ Royal London Hospital test					

Sn = Sensitivity; Sp = Specificity; LR – = Negative likelihood ratio; LR + = Positive likelihood ratio

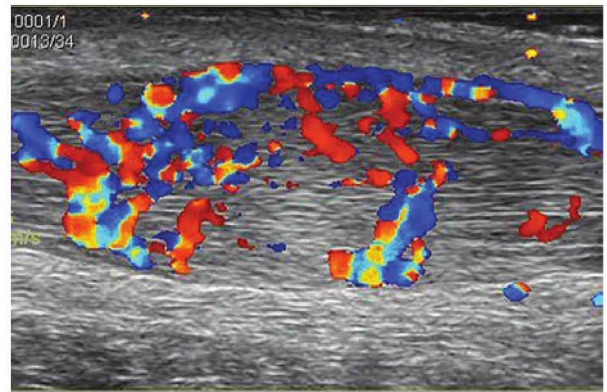
Blue – Large change in likelihood (LR – <0.1, LR + >10)

Red – Rarely important change in likelihood (LR – >0.5, LR + <2.0)

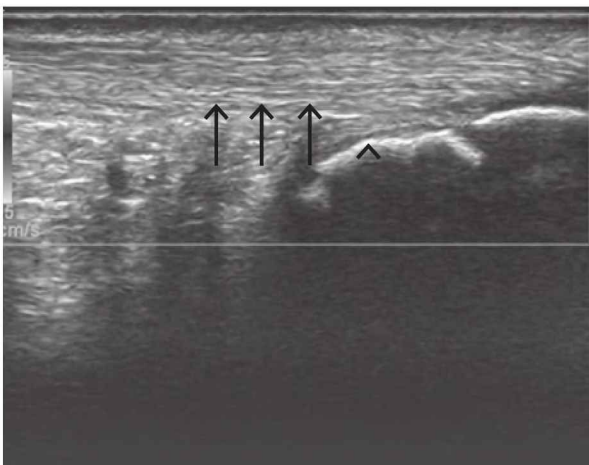
* Positive likelihood ratio cannot be calculated when specificity equals 100



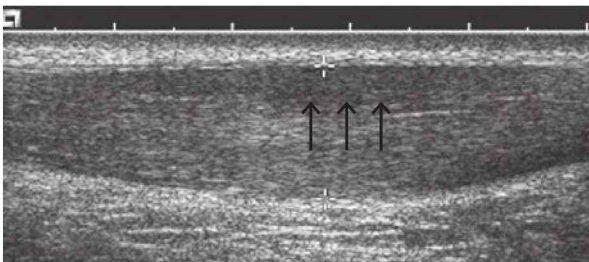
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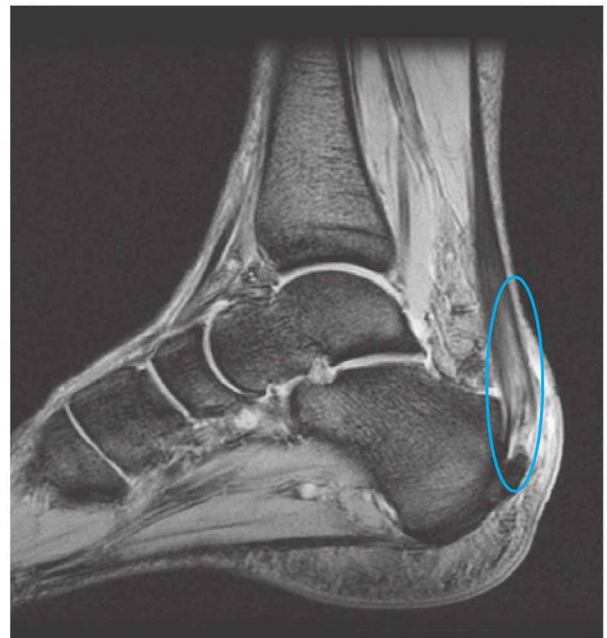
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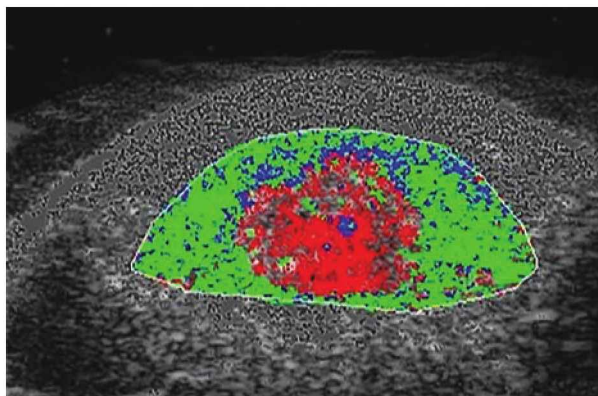


(c)



(e)

Figure 40.4 Imaging findings in the Achilles region (a) Calcification is a common finding in pathological tendons. It should not influence management and a bone spicule (circle) is very difficult to find in surgery. The take-home message is to manage according to symptoms and function; do not manage the radiograph (b) Greyscale ultrasound of a normal Achilles tendon (arrows). The superior calcaneum (arrowhead) prevents sound waves from penetrating, hence the acoustic 'shadow' deep to the calcaneal margins (c) Greyscale ultrasound of an Achilles tendon with mild morphological abnormality (arrows)—thickening of the tendon and less echodensity (because of increased matrix ground substance and associated fluid) (d) Colour Doppler ultrasound showing abnormal vessels in symptomatic tendinopathy. Appearances do not mirror symptoms enough to provide definitive clinical support (e) MRI appearance of increased signal in the distal Achilles tendon extending into the insertion (blue oval)



(f)

Figure 40.4 (cont.) (f) Ultrasound tissue characterisation (UTC). This ultrasound modality has been most used in Achilles tendinopathy as the subcutaneous tendon anatomy is well suited to the linear probe. Whether the instrument will have widespread clinical utility remains to be seen

Table 40.4 Imaging findings in overuse Achilles tendinopathy (i.e. not a complete rupture). Imaging appearances do not equate with symptoms nor do they predict time to return to sport (recovery). See also Figure 40.4

Imaging appearance	Variations seen in clinical practice
Normal appearance on ultrasound/MRI	It is possible, but unusual, to have a normal imaging with symptoms of Achilles tendinopathy. Carefully evaluate alternative structures as a potential source of pain
Ultrasound and MRI tendon swelling	Tendon thickening varies, but there is often a fusiform swelling. Evaluate contribution of peritendon to swelling
Ultrasound discontinuity of tendon fascicles	Hypoechoic regions are often seen inside the thickened tendon. This corresponds to a fluid-rich region with irregular collagen fascicle arrangement
Doppler ultrasound evidence of high blood flow	May be extensive or absent and may vary in amount when imaged after rest or activity, and on different days
Increased MRI signal	Abnormal high signal, best seen on T2-weighted sequences (see also Chapter 15 for more advice for clinicians on interpreting MRI)

Inflammatory cells are few in midportion Achilles tendinopathy.^{26, 30} Intratendinous microdialysis^{31, 32} and analysis of appropriately prepared biopsy tissue using cDNA-arrays and real-time quantitative polymerase chain reaction (PCR)³³ all failed to show prostaglandin-mediated inflammation.³⁴ The debate around whether Achilles tendinopathy has an inflammatory component or not³⁵ is complicated because of the various definitions of inflammation.³⁶

Biopsy specimens from midportion Achilles tendinopathy show increases in some cytokines and neuropeptides, such as substance P and calcitonin gene-related peptide (CGRP). Activated tendon cells can produce substances that can cause pain that may be detected by nerves in the peritendon.³⁷ Tendon pain is complex and the source is often unknown.³⁸ (See Chapters 5 and 6 for more on pain biology.)

Predisposing factors for Achilles tendinopathy

Midportion Achilles tendinopathy is most common among active people, but it also occurs in inactive individuals. Irrespective of the baseline level of activity, relative tendon overload is the main trigger. This may occur in a single episode or, more often, over a period of time. Athletes, for example, may overload their tendon during an intensive period of competition or training, whereas for inactive people the commencement of a new exercise regimen may be sufficient. Although overload is considered to be the primary inciting event for tendinopathy, tendons' response to load is also influenced by a range of intrinsic and extrinsic risk factors.

Intrinsic risk factors

Modifiable intrinsic risk factors include central adiposity in men and peripheral adiposity in women,³⁹ increased BMI,^{40, 41} hypercholesterolaemia and diabetes mellitus (type 1 and 2).^{42, 43} Non-modifiable factors include previous injury/tendinopathy,^{44, 45} genetic predisposition (certain genes coding for collagen),^{46, 47} exposure to fluoroquinolones⁴⁸ or steroids,⁴⁹ and (for insertional tendinopathy) inflammatory arthritis.⁵⁰ Older age may also be a risk factor; however, the current evidence is conflicting, suggesting that other specific factors such as cumulative load may influence this association.⁵¹

There are also a number of modifiable biomechanical factors associated with Achilles tendinopathy, shown in Table 40.5. These either increase strain on the tendon or reduce its ability to bear load. Contrary to popular belief, there is little evidence of an association between subtalar joint pronation and Achilles tendinopathy.⁵² In fact, a *supinated and rigid foot* is associated with tendinopathy.⁵³⁻⁵⁵

Extrinsic risk factors

In active individuals, the main extrinsic risk factor for Achilles tendinopathy is a recent change in tendon loading, such as increasing distance, speed or frequency of training/competition with insufficient recovery between loading periods.⁶¹⁻⁶⁴ In particular, a sudden increase in activities involving high energy storage, such as jumping and landing, is thought to be important. In sedentary individuals this stimulus may be significantly less but the principles still apply. Other extrinsic risk factors such as changes in footwear or training surface may be important, especially in elite athletes, but are secondary to load.

Treatment of midportion Achilles tendinopathy

Experienced clinicians begin conservative treatment by identifying and correcting possible aetiological factors. Most importantly, this involves management of the patient's tendon load (first decrease load to control symptoms and then increase load to improve tendon capacity). Biomechanical limitations such as restricted ankle dorsiflexion and hallux rigidus should also be addressed if possible. Below, we outline common treatments for Achilles tendinopathy including targeted exercises (mechanotherapy,⁶⁵ see Chapter 17), injections and surgery.

Focused rehabilitative exercise training: the hub of tendinopathy treatment

Exercise training is at the hub of effective treatment for midportion Achilles tendinopathy (Fig. 40.5). Such

treatments became the preferred non-surgical management for Achilles tendinopathy after Alfredson published his eccentric loading program in 1998.⁶⁶ Repeated and, initially, pain-provoking exercise led to patients being pain-free in about 12 weeks. This program was a landmark in moving treatment away from rest and passive approaches. In addition to the good clinical results, ultrasound and MRI follow-up demonstrated that patients' tendons improved in appearance and decreased in thickness, although they did not return to a normal appearance.⁶⁷

PRACTICE PEARL

Various types of exercise programs are effective for tendinopathy—there does not appear to be anything 'magical' about eccentric exercise in this setting.^{68, 69} The combination of eccentric and concentric movements may be as beneficial as eccentric movements alone.⁶⁹⁻⁷¹

Doing both concentric and eccentric exercises may also promote muscle strength and endurance adaptation.^{69, 72} In addition, the inclusion of more dynamic exercise may not adversely affect the outcome.⁷³

The take-home message in 2017 appears to be that progressively loading the muscle tendon unit (kinetic chain) to promote recovery of full-capacity tendon is likely the effective stimulus; this argues against 'recipe

Table 40.5 Biomechanical factors that are associated with Achilles tendinopathy and their possible mechanisms

Factor	Possible mechanisms	Assess by
Relative weakness in plantarflexion strength ⁵³	Increased load on tendon by failing to attenuate load	Single-leg heel raise, single-leg hop, dynamometer or strength/endurance tests (heel raise to fatigue)
Increased dorsiflexion range of motion ⁵³	Increased tendon load (via longer loading period) and longer levers Possible collagen abnormalities (e.g. benign hyperlaxity syndrome/Ehlers–Danlos syndrome)	Weight-bearing (knee to wall) and non weight-bearing assessment of dorsiflexion ROM
Reduced dorsiflexion range of motion with or without supinated foot type ^{54, 56}	Increased tendon load (due to faster loading time)	Weight-bearing (knee to wall) and non weight-bearing assessment of dorsiflexion ROM
Reduced knee flexor strength ⁵⁷	Increased tendon load by increasing demand on plantarflexors	Dynamometry or clinical strength/endurance tests (bridging) ⁵⁸
Poor quality tendon structure ^{59, 60}	Reduced capacity to tolerate load Less effective load-bearing tissue	Ultrasound, Ultrasound Tissue Characterisation (UTC), MRI

physiotherapy' ('Do X of these exercises twice a day like a robot'). The 'capacity' concept⁷⁴ is illustrated and discussed in Chapter 4 (see Fig. 4.14). That key diagram can be photocopied and marked up to show patients their current capacity (triangle at bottom left of the figure) and their required capacity (triangle at top right of the figure). Different exercises are prescribed to improve the patient's capacity (Chapter 4, four-stage training program).

Four-stage, progressive Achilles exercise program adapted to the clinical assessment

In this chapter, the broad 'capacity' concept is customised for Achilles tendinopathy in Figure 40.6. The four stages of treatment include three relating to developing tendon capacity and one that provides analgesia (isometric exercises). The four stages are outlined in the box.

Prevention: Can exercises prevent tendinopathy?

Eccentric exercise appears to be ineffective as a prophylactic treatment to prevent Achilles tendinopathy. Soccer players who performed eccentric exercise and stretches during the playing season were no more likely to be free of Achilles tendon injury than those who did not.⁵⁹

Beware: Is the plantaris tendon a culprit?

The plantaris tendon is usually a tiny benign tendon medial to and removed from the Achilles (Fig. 40.8a) In some people, it enlarges and compresses the Achilles tendon, in others it can be ensheathed in the Achilles peritendon (Fig. 40.8 b, c, d) In both cases the variable properties of the plantaris and Achilles cause compression of the Achilles tendon and may provoke tendinopathy.⁷⁷⁻⁷⁹ The plantaris was first identified as a potential contributor to midportion Achilles region pain around 2011.⁸⁰ The clinical clues to this culprit are pain and tenderness that are more medial and more proximal to the 'classic' midportion Achilles tendinopathy symptoms and pain aggravated by loading in dorsiflexion.

If this diagnosis is suspected, imaging (ultrasound or MRI) is indicated.

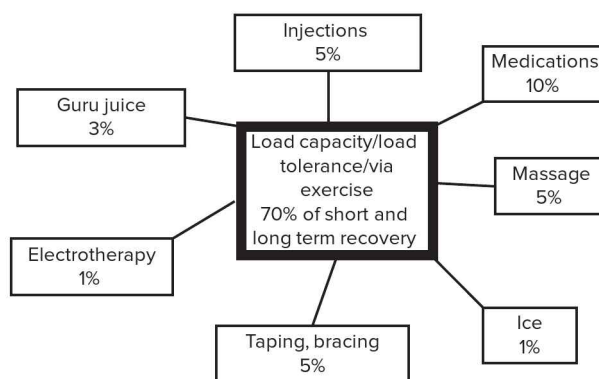


Figure 40.5 Treatment of midportion Achilles tendinopathy. Warning! Warning! The numbers in this conceptual figure are not to be taken literally. There are many possible treatment options for Achilles tendinopathy. Systematic reviews indicate that various forms of exercise prescription provide superior pain relief over control treatments (level 1 evidence). The relative effect sizes suggested in this figure should be considered estimates and will vary from patient to patient. The estimates of relative importance reflect the chapter authors' expert clinical experience (level 5 evidence). It highlights that not all treatments are created equal

Plantaris management

Conservative management of this condition follows the loading principles of the four-stage program (outlined in the box below, Figs. 40.6 and 40.7) while reducing compression of the plantaris against the Achilles tendon. This is achieved by ensuring the patient completes all exercises with a moderate degree of ankle plantarflexion, and avoids loads in dorsiflexion, especially in the early stages of rehabilitation. Using a high heel raise similar to those used for insertional Achilles tendinopathy (see below) will allow the patient to complete these exercises pain-free. At stages 3 and 4 of the rehabilitation program, it is appropriate to gradually reduce the heel height until the exercise can be completed through the full range.

The four-stage program for midportion Achilles tendinopathy

This four-stage rehabilitation program incorporates the exercises from various published programs and is clinically effective for Achilles tendinopathy. The key elements are its (i) stages, (ii) progressive nature, and (iii) flexibility (to adapt it to clinical re-assessment).

Stage 1—Pain relief. Isometric exercises

Patients focus on isometric calf raise exercises that relieve pain. These can be done with simple body-weight holds (single- or double-leg depending on strength and pain), but athletes require weights in a standing heel raise machine.

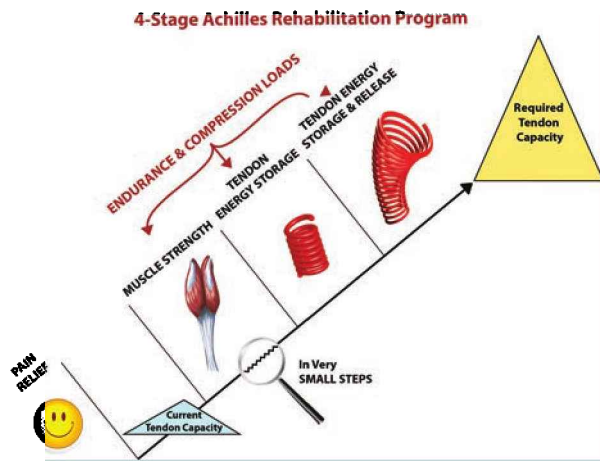


Figure 40.6 The four-stage program for midportion Achilles tendinopathy. Pain relief (with isometric exercise) allows the patient to undertake exercises to improve tendon capacity (see also Fig. 4.14)

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The isometric holds should be long (up to 45 seconds) with the load that can be held without perturbation in the muscle. Four to five repetitions interspersed with 2 minutes rest have proven effective in our clinical experience. This exercise can be repeated several times a day to provide the patient with pain relief as needed.

In those who have substantial pain and pathology, the seated heel raise exercise offers the opportunity to use loads that are below body weight; it is possible to find an isometric load that is suitable for even the most compromised tendon.

Stage 2—Isotonic strength endurance where appropriate

Progress the patient to this stage when Achilles tendon pain has settled and the patient is competent at isometrics and slowly building both (i) hold time (45 s, as above) and (ii) weights (adjusted based on what the patient can do and hold steady). Isotonic exercise (slow concentric and eccentric heel raises) with body weight or additional weights with a weight-based machine strengthens muscle and provides beneficial load to tendon. Aim to go from full plantar flexion to plantar grade (i.e. neutral). Only lower into dorsiflexion for athletes who need strength in this range.

PRACTICE PEARL

Remember: Isometrics can be done prior to isotonic loading as they may relieve cortical motor inhibition that will assist with strength gains.^{75,76}

Ensure the patient reaches full range (maximal) plantar flexion without using trick movements such as supination. When is strength adequate for progress? High-functioning patients will be able to raise (using a single leg) up to 1.5 times their body weight as additional load; patients whose demand is lower can progress when they can raise (using a single leg) the equivalent of their own body weight as additional load. Be sure to train endurance with high-quality movement (no trick movements) before progressing to stage 3. There should be evidence of muscle bulk returning at this point. Stage 2 exercises must be maintained at least twice a week for the remainder of the person's active life.

If necessary, a strength endurance exercise can be added, where one set to fatigue of repeated heel raises is completed each night. A functional alternative is stair climbing with the foot maintained in sustained plantarflexion; that is, the heels do not touch the steps while climbing.

Stage 3—Energy storage exercises

Exercises in this stage place faster loads on the tendon and this causes the tendon to store energy and begin to act as a spring. In stages 3 and 4 (see below) the patient does not carry additional weight; body weight is sufficient and provides close to maximum load for the tendon. Any fast exercise, particularly if it includes eccentric loading, will induce energy storage, so exercises such as slow skipping-type activities and faster stair climbs are ideal. Stage 3 and 4 loads should be completed a maximum of three times a week. The remaining days can be used to continue strength work and cross-training.

Stage 4—Energy storage and release exercises

These exercises are a progression of stage 3, where the speed increases and the release of energy from the tendon occurs. Faster skipping, running drills and change of direction drills begin to prepare the athlete for return to training.

Return to training and competition—as after all injuries

This is an essential, and often neglected, element of successful tendon rehabilitation (see also Chapters 18 and 19). We distinguish this final element from the four-stage tendon rehabilitation so as not to dilute the tendon-specific elements of that program. Return-to-training or return-to-play exercises are constructed with the athlete's specific functional requirements in mind. Thus, there is no recipe because the demands of an elite track athlete and recreational runner are very different even within the one activity/sport (running). Transition to training is permitted when the person can complete stage 4 exercises repeatedly without an increase in symptoms the morning after the rehabilitation session.

How does exercise promote tissue repair in tendinopathy?

Exercise therapy can affect any tissue at the cellular level through the process of mechanotherapy,⁶⁵ and this is detailed in Chapter 17.

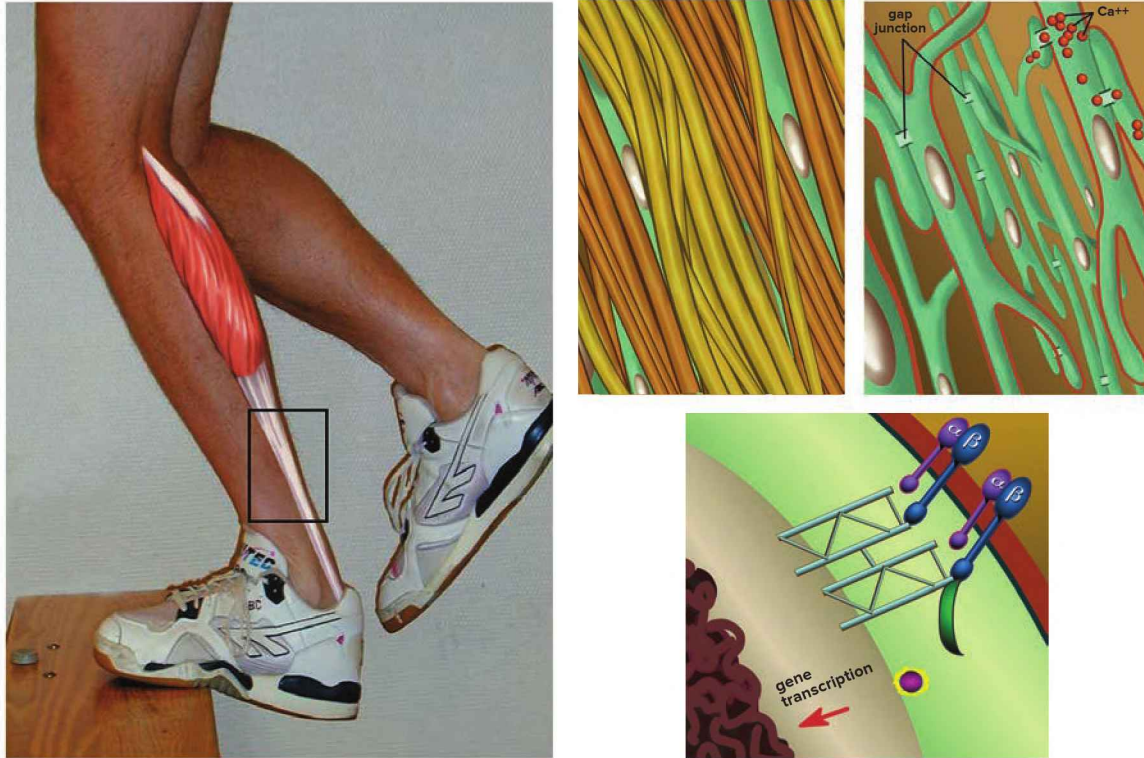


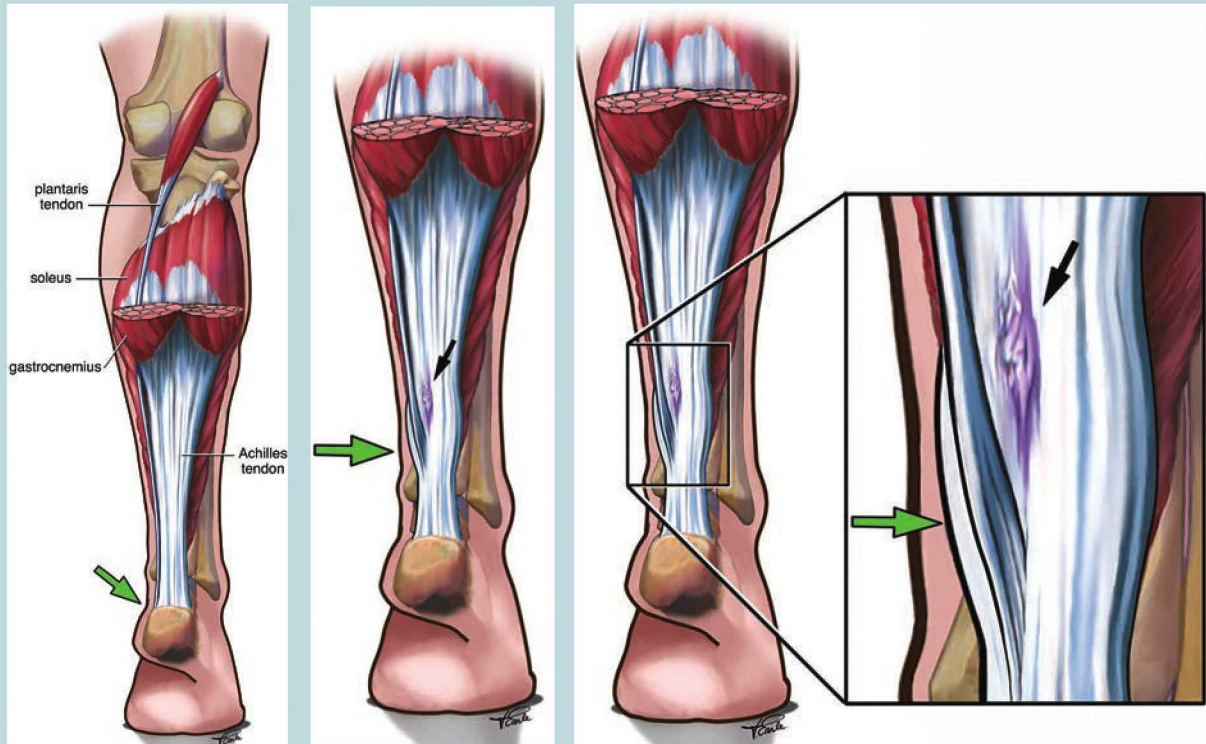
Figure 40.7 Loading exercises (left panel), cause sliding of collagen fibres (illustration of the microscopic view, top centre panel) leading to intercellular⁶⁵ communication via gap junctions (top right panel), and intracellular communication with the cell nucleus (lower right panel).

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Because the aberrant plantaris can be fixed to the Achilles tendon (Fig. 40.8), conservative management may not be successful. Sweden's Professor Håkan Alfredson was a pioneer in identifying this pathology in patients with painful Achilles tendons.⁸⁰ Surgery is performed under ultrasound and Doppler guidance. The symptomatic pathological region is approached using a medial incision and the adherent tendons are separated or the plantaris tendon is extirpated (divided and removed).⁸¹

Other interventions used in conjunction with staged progressive exercise therapy

We summarise interventions that have been used in conjunction with an exercise program (Table 40.6) and note two things. First, exercise is excellent for tendons and, second, added interventions add minimally to exercise therapy. It is important to apply the 'exercise is medicine' message from public health to tendons.



(a)

(b)

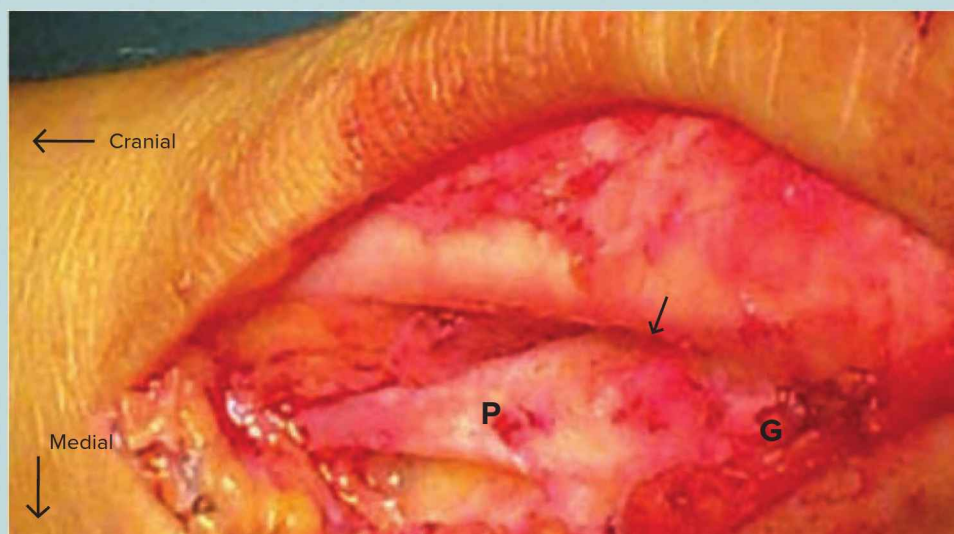
(c)

Figure 40.8 The anatomical relationship between the (a) Anatomy considered ‘normal’ for the Achilles and plantaris tendons (b) The plantaris commonly inserts in an aberrant manner—to the medial border of the Achilles tendon (green arrow). This can promote Achilles tendinopathy (black arrow) (c) Cross-sectional schematic showing commonly seen variations in the anatomical relations

It is unfortunate that among many patients and clinicians, exercise is incorrectly assumed to be less effective than traditional medical therapies such as medication or injection.

Injections

There are a range of substances that have been used both in and around the Achilles tendon. The basis for injection is variable, but should only be used when there is failure



(d)

Figure 40.8 (cont.) (d) Plantaris tendon (P) adherent to the medial aspect of the Achilles tendon (arrow) with prominent richly vascularised fat tissue that (G) contains sympathetic and sensory nerves

to respond to a good quality exercise program for an extended time.

Corticosteroid injections

Corticosteroid injections are falling out of favour for long-term management of tendinopathy.⁸⁶ Corticosteroid has an immediate and short-term pain-relieving effect (for up to 8 weeks). The mechanism for cortisone potentially healing a tendon has not been clearly articulated and clinical trials of the medication consistently point to poorer long-term outcomes than 'wait and see' in various tendons. There has been no high-quality study of corticosteroid injection versus placebo or 'wait and see' at the Achilles tendon as there has been at the elbow. Longer-term outcomes at various anatomical sites are poorer than both no intervention and exercise interventions.⁹⁶

Autologous blood and platelet-rich plasma (PRP)

With ROBERT-JAN DE VOS

Autologous blood and platelet-rich plasma (PRP) has become popular in recent years. They have been proposed to improve tendon healing because of the growth factors and cytokines present in blood.^{97, 98} There are many protocols for the preparation and injection of the blood products and cells, making it harder to compare studies. A

number of clinical trials have been performed on patients with chronic midportion Achilles tendinopathy. One small randomised study did not show benefit of PRP compared to eccentric exercise therapy.⁹⁹ One placebo-controlled randomised trial on the effectiveness of PRP combined with eccentric exercises did not show differences in clinical recovery or improvement of tendon structure.^{19, 100}

Sclerosing

This treatment consists of injecting a vascular sclerosant (polidocanol, an aliphatic nonionised nitrogen-free substance with a sclerosing and anaesthetic effect) using ultrasound guidance in the area of neo-vascularisation anterior to the tendon. In a small double-blind, randomised controlled study comparing the effects of injections of a sclerosing and a non-sclerosing substance (lidocaine plus adrenaline), the sclerosing therapy had superior results to placebo.⁸³ In a 2-year clinical and ultrasound follow-up, good clinical results persisted and the tendon had decreased in thickness.⁸⁴ Whether the sclerosant decreases vascularity or pain is not known. Sympathetic and sensory nerves run alongside blood vessels.¹⁰¹ Short-term follow-up of tendons that have had vessels sclerosed demonstrated an increase in both short- and medium-term vascularity.¹⁰² This, in combination with

Table 40.6 Best available evidence for treatments of midportion Achilles tendinopathy

Intervention	Level of evidence	References
Mechanotherapy		
Eccentric exercise	1	66, 69, 82
Isotonic concentric/eccentric exercise	1	69, 73
Isometric exercise	5	76
Injections		
Sclerosing	2	83, 84
Platelet-rich plasma (PRP)	1	19
Corticosteroid	1	85, 86
High volume imaging guided (HVIG)	2	87
Cell injections	(Rabbit model)	88
Electrotherapy		
Shockwave therapy	1	89, 90
Laser	1	21, 91, 92
Surgery		
Ultrasound-guided scraping	2	93, 94
Open surgical revision	4	95

Blue = use supported by meta-analyses (systematic reviews) or randomised controlled trials

Orange = use supported by non-randomised studies and case series

Black = use supported by expert opinion or clinical guidelines

Red = current evidence does not support common clinical use

the anaesthetic properties of polidocanol, suggests that a change in vascularity is not necessary for a change in pain. Complications include a small risk of rupture and infection; they are very rare if small volumes of polidocanol are used.

High-volume imaging-guided (HVIG)

Despite the general lack of success of injection therapies for Achilles tendinopathy, optimists continue to trial various combinations of compounds in low-quality case series. One combination, called high-volume image-guided injection (HVIGI), includes normal saline, local anaesthetic and corticosteroid. Uncontrolled case series have reported decreases in patient pain over time and improvements in VISA-A scores.⁸⁷ A 2016 study acknowledged fatal (in the research sense of that word) confounding in study design; investigators were unable to draw conclusions from a retrospective series using high-volume injection.¹⁰³ As summarised in Table 40.6, current evidence does not support such treatments. One can always speculate as to possible mechanism of action for any physical agent

(volume) or compound (pharmaceutical or naturopathic). See Chapter 2 for discussion of 'evidence' in contemporary medical practice.

Cell-based therapies

Therapies that inject cells (stem, fibroblasts) into the tendon are nascent but currently only supported in humans by case series.¹⁰⁴ It is critical to remember that tendon pathology is largely a pathology of hyper-cellularity, so the underlying premise for the use of additional cells is not obvious. Most biological processes need a careful interplay of elements (cells, hormones, cytokines). Because the ideal tendon milieu for healing has not been identified, it appears that a lot more research will be needed before clinicians can customise cell therapy (biologics) to match the healing stage of the underlying tendon. Inventors and investors stand to make large profits in the biotechnology arena. It is also important to be explicit that some healthcare systems provide financial incentives (fee for service) for clinicians to offer permitted, but unproven, treatments.

Medications

Nonsteroidal anti-inflammatory medications have failed to improve patient outcomes in tendinopathy. This is supported by individual randomised controlled trials (RCTs), one systematic review¹⁰⁵ and clinical impression. Leading SEM physicians do not prescribe NSAIDs for long-term treatment of tendinopathy in the routine clinical setting. They may consider these medication as an adjunct for short-term pain relief or if a player continues to compete with tendinopathy because of a short-term goal.

Professional sport is a different context and players often seek the analgesic effect of NSAIDs.¹⁰⁶ The ethical issues of using pain-relieving medication in sport are beyond the scope of this chapter.

Electrophysical agents

There is evidence that extracorporeal shock wave therapy (ESWT) is effective in treating lower limb tendinopathies including Achilles tendinopathy (AT).^{89, 107-109} This mechanical treatment is described in Chapter 17. Studies have shown small benefits over and above eccentric exercise in treating midportion tendinopathy.⁸⁹ The mechanism of action of ESWT is unknown, but it may affect the nerves or provide a physical stimulus to tendon cells. The modality caused disorganisation of tendon collagen in Shetland ponies.¹¹⁰ It damages renal tissue when used in humans for lithotripsy. Because it is a physical agent, collateral damage is inevitable. Whether this damage is clinically relevant in human tendons has not been studied.

One RCT found that laser therapy was an effective adjunct to eccentric exercise compared to a placebo.⁹¹ Placing laser therapy in a clinical context is difficult as how it has an effect on tendon pathology and pain is unknown.

Surgical treatment

A systematic review of Achilles tendon surgery identified 23 studies with almost 1200 patients in total who underwent five different categories of types of surgery.¹¹¹ The main types of surgery, and those common in clinical practice, were open surgical debridement (11 studies) and minimally invasive procedures (7 studies). Patient satisfaction and complication rates were the main reported outcomes.

PRACTICE PEARL

It is difficult to draw conclusions about Achilles tendinopathy surgery because no study has yet (as of November 2016) compared surgical intervention with non-surgical or placebo intervention.

The minimally invasive procedures yielded lower complication rates with similar patient satisfaction in comparison with open procedures.¹¹¹ In clinical practice, it is clear that Achilles tendon surgery is not a quick fix for patients.

If a patient is not making progress with an exercise program, it would be reasonable to refer the patient to a fellow clinician who is expert in treating tendinopathy (e.g. specialist physiotherapist, sport and exercise physician, or other non-surgical specialist) rather than opting to refer to a surgeon who is not expert in treating tendinopathy. There are of course, expert sports surgeons who provide excellent advice (Chapter 1). The key is to find tendon-specific expertise from among the disciplines in our community.

INSERTIONAL ACHILLES TENDINOPATHY INCLUDING RETROCALCANEAL BURSTITIS—THE 'ENTHESIS ORGAN'

There has been confusion regarding the terminology in this region (see box). To aid new clinicians, we first consider three separate diagnoses: (i) insertional Achilles tendinopathy, (ii) retrocalcaneal bursitis and (iii) Achilles (superficial calcaneal) bursitis. In clinical practice the first two commonly occur together because they have a shared aetiology—excessive tendon load in compression against the calcaneum.

The key concept is that at the Achilles insertion, the tendon, bursa and bone are intimately related so that in dorsiflexion (compression) they operate in partnership to *reduce* strain on the insertion itself¹¹²—the 'enthesis organ'¹¹³ (Fig. 40.9). Excessive dorsiflexion, especially with high tensile loads, will predispose to mechanical irritation of the bursa and the tendon just proximal to the insertion.¹¹⁴ In response to such irritation, the tendon adapts with increased fibrocartilage and, if the loads are excessive, tendon pathology. If the compression is sufficient and over an extended time period, the tendon may develop foci of ossification. This is how the body responds to high compressive loads.

Anatomy and the key role of compression

At the Achilles tendon insertion, the fibrocartilaginous walls of the retrocalcaneal bursa extend into the tendon (Fig. 40.9b) and the adjacent cartilage-covered calcaneum. Let us examine each element in turn. First, the calcaneum and the 'Haglund deformity'.

Historically, much has been made of the so-called Haglund deformity (see box below). However, this is normal anatomy, the way the greater trochanter serves as a cam for the gluteal tendons and the ischial tuberosity plays that role for hamstring tendons.^{117, 118} Tendons insert at an angle into a hollow over a bony prominence to obtain mechanical advantage (i.e. a cam) and to protect the insertion from high loads. This is

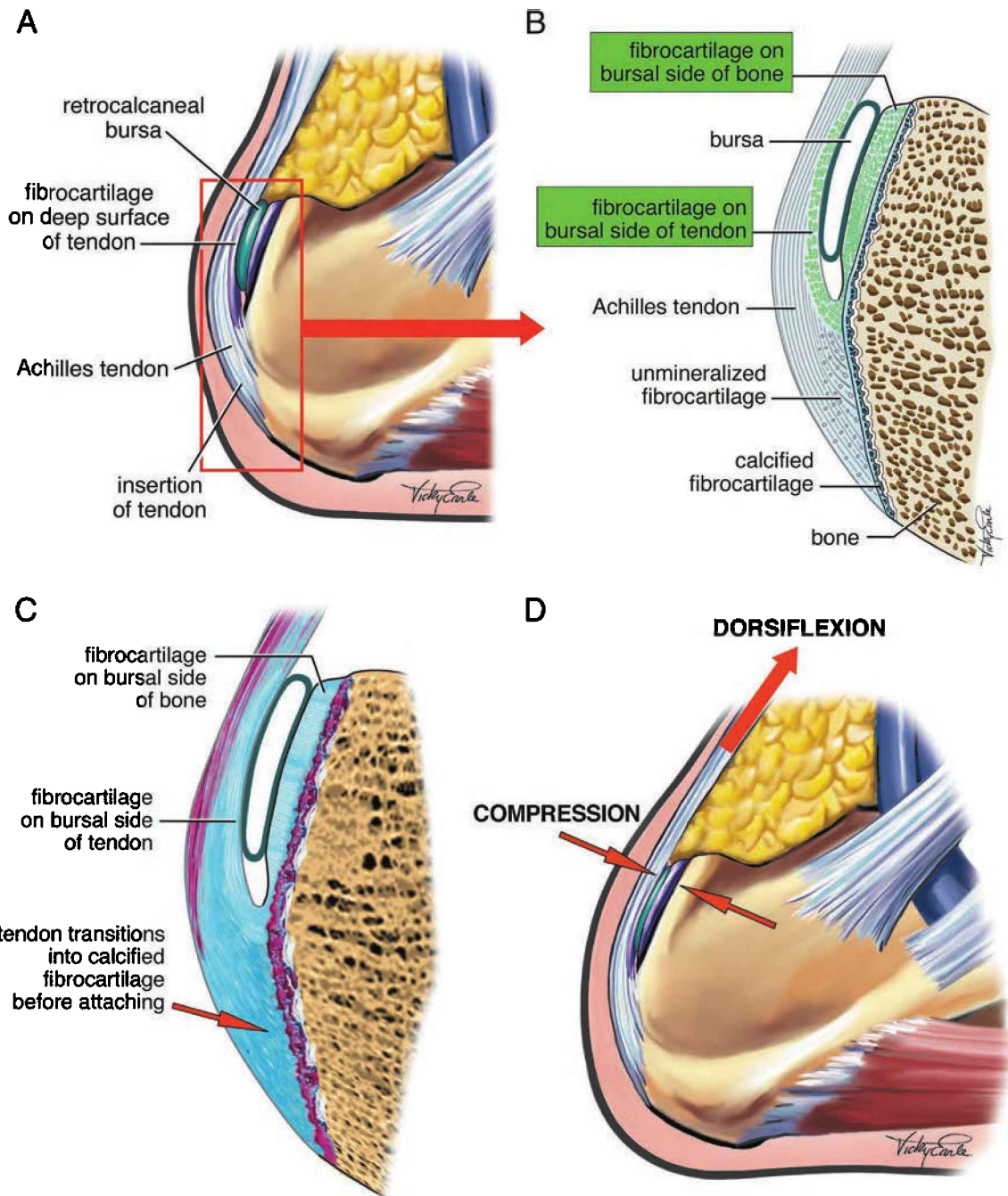


Figure 40.9 The critical role of compression in 'retrocalcaneal bursitis' (a) The anatomy represents an 'enthesis organ'. A prominent calcaneum greatly predisposes to mechanical irritation of the bursa and development of excess fibrocartilaginous tissue on both sides of the bursa (b). Note that fibrocartilage is normal at that site, but it can become thickened with excessive compression (c) Shows the microscopic appearance of the tendon insertion (red and blue, [from photomicrograph]) attaching to the artist's rendition of the calcaneum (d) This image highlights the effect of dorsiflexion—excessive compression. This underpins the treatment principle of limiting dorsiflexion (compression) with a substantial heel raise

Haglund: time to take this eponym out of the books?

There has always been considerable confusion about the terminology associated with pain in this region. Originally, Haglund described pain in the hindfoot caused by a prominent posterosuperior corner of the calcaneus in combination with wearing a rigid low-back shoe.¹¹⁵ Since then, various terms have been used incorporating Haglund's name.

'Haglund-type calcaneus' or 'Haglund deformity' or 'Haglund's exostosis' is a descriptive label for a protuberance of the posterolateral calcaneum, where a calcaneal prominence can often be palpated. This is also known as 'pump bump', with alternative names such as calcaneus altus, high prow heels, knobby heels, and cucumber heel. This deformity is frequently present in asymptomatic patients. There is a school of thought that considers it normal and thus better referred to as a Haglund *morphology*.

Haglund's syndrome involves a painful swelling of an inflamed retrocalcaneal bursa, sometimes combined with insertional tendinopathy of the Achilles tendon. It must be distinguished from Haglund's disease, which is the term for osteochondrosis of the accessory navicular bone.

It has been proposed¹¹⁶ that terms such as Haglund's disease, Haglund's syndrome, Haglund's deformity, pump bump, calcaneus altus, high prow heels, knobby heels and cucumber heel no longer be used.

the point where the tendon is compressed against the calcaneus in dorsiflexion; this load can induce pathology at this site.

Much of the pathology is seen at the upper calcaneal/bursa/tendon interface (the cam), where the compression of the tendon against the upper calcaneus occurs.

Clinical assessment

Good clinical practice includes evaluation of the tendon and tendon insertion by taking a careful history and performing functional tests in addition to inspecting and palpating the region for bony prominence and local swelling. Discriminating between pain with tensile, compressive and combined loads can help with diagnosis. For example, hopping on the toes (tensile loads) may give some pain, but a hop-lunge into dorsiflexion (compression and tensile loads) may be more painful at the insertion.

PRACTICE PEARL

A heel raise of 30–50 mm is a good practical way of reducing compressive loads.

Ultrasound and MRI can help to assess the extent of pathology in the tendon and the bursa. Ultrasound + Doppler (US + DP) examination can image the bursa and tendon insertion (thickening and localised high blood flow).

Symptoms of insertional Achilles tendinopathy, as with any enthesopathy, should raise suspicion about the possibility of rheumatoid arthritis or spondyloarthropathy, and this is particularly true if symptoms are bilateral.

Treatment

Treatment must consider the enthesis organ as a unit. Isolated treatment of a bursitis is generally unsuccessful. Their key principle is to use a staged loading program

with particular attention to avoiding dorsiflexion (to avoid compression of the tendon).¹¹⁹

Repeated stretching is not recommended as it provides compression load to the insertion.

Alternative treatments for chronic insertional tendinopathy include sclerosing of regions with high blood flow and nerves with polidocanol.^{120, 121}

Treatment of insertional tendinopathy with extracorporeal shock wave therapy (ESWT) showed more benefit than a low-quality exercise,¹²² but the eccentric program used was designed for mid-Achilles tendinopathy, and better benefits from exercise are evident when using the specific insertional protocol.¹¹⁹

PRACTICE PEARL

Conservative management is usually successful, while surgery can require long-term rehabilitation. Our clinical experience includes elite middle-distance runners who were unable to return to their previous level of competition after surgery. Some case series report unflattering outcomes.¹²³

Only a few case series have reported outcomes after surgery for insertional Achilles tendinopathy.^{124–126}

Retrocalcaneal bursitis

The reader coming to this subheading 'cold' from the index will also need to read the section above. Retrocalcaneal bursitis is not generally an isolated condition in the sports medicine setting. Inflammation of the retrocalcaneal bursa (Fig. 40.9a) results in a painful soft tissue swelling, medial and lateral to the Achilles tendon at the level of the posterosuperior calcaneus.

PRACTICE PEARL

Histopathologically, the fibrocartilaginous bursal walls show degeneration and/or calcification, with hypertrophy of the synovial infoldings and accumulation of damaged fibrocartilaginous cells and fluid in the bursa itself.¹²⁷

B

Rarely, but not to be missed, this condition may signal an underlying infectious bursitis due to an inflammatory arthropathy.¹¹⁶

To determine whether tendon or bursa is the primary pain generator, one of the clues suggesting retrocalcaneal bursitis is irritable pain and symptoms on low-load activities such as heel raises or end-range compression in dorsiflexion. A lot of fluid or thickening of the bursa on imaging (ultrasound or MRI) may be associated with the bursa being a pain generator. Conversely, a little fluid on imaging is normal in active populations.

For treatment of the combined ‘enthesis organ’—the bursa and associated tendon—see the section above and note the need to avoid compression (see Fig. 40.9). Principles of quality tendon management apply (see the four-stage Achilles rehabilitation program above).

Injection of corticosteroid into the bursa, preferably under ultrasound guidance, can be effective in relieving symptoms¹²⁸ but is associated with an increased risk of Achilles tendon rupture.¹²⁹ This may be due to the presence of a connection between the retrocalcaneal bursa and the anterior fibres of the Achilles tendon.

If conservative treatment fails, then surgery to remove the bursa and any associated bony prominence is often advocated.^{123, 130} Our reasons for making that an absolute last resort have been laid out.

Achilles (superficial calcaneal) bursitis

The Achilles or superficial calcaneal bursa (Fig. 40.1b) is located between a calcaneal prominence or the Achilles tendon and the skin, resulting in a visible, painful, solid swelling and discolouration of the skin. It is most often located at the posterolateral aspect of the calcaneus. Histopathologically, the subcutaneous bursa is an adventitious bursa, which is acquired after birth, and develops in response to friction. Unlike the retrocalcaneal bursa, which is lined by fibrocartilaginous cells, this bursa consists of hypertrophic synovial tissue and fluid. A superficial calcaneal bursitis can be further specified by its location; that is, posterior, posterolateral or posteromedial.

It is generally caused by excessive friction, such as by heel tabs, or wearing shoes that are too tight or too large. Various types of stiff boots (skating, cricket bowling, etc.)

can cause such friction. Removing the source of the friction by using a punch to widen the heel of the boot, or providing ‘donut’ protection to the area of bursitis as it resolves is generally successful. As with many conditions, corticosteroid injection was considered a quick fix in the 1990s but is falling out of favour as a definitive treatment. The underlying cause of the pathology (the culprit) needs to be addressed.

POSTERIOR IMPINGEMENT SYNDROME

with SUSAN MAYES

Posterior impingement syndrome of the ankle refers to impingement of soft tissue or bony structures between the posterior aspect of the tibia and the superior aspect of the calcaneus in extremes of plantarflexion. Gradual onset of posterior pain may arise from impingement of an enlarged posterior tubercle of the talus (Fig. 40.10a) or an os trigonum (Fig. 40.10b), but is usually the result of impingement of the synovium or capsuloligamentous structures. This condition is common among ballet dancers, gymnasts, cricket bowlers and football players, all of whom maximally plantarflex their ankles in a loaded position. It is also seen secondary to ankle plantarflexion or inversion injuries. It can also arise from an acute presentation—when a forceful hyperplantarflexion fractures the posterior process of the talus (Chapter 41).

The diagnosis of posterior impingement syndrome is suggested by pain at the posterior aspect of the ankle and confirmed by a positive posterior impingement test—pain reproduced on passive plantarflexion of the ankle (Fig. 40.3f). Flexor hallucis longus (FHL) tenosynovitis commonly coexists with posterior impingement in dancers (Chapter 42). Pain with active first-toe flexion and extension in full ankle plantarflexion indicates FHL tenosynovitis (Fig. 40.3e). Another differential in ballet dancers is Achilles tendinopathy. The posterior impingement test is pain-free (negative) in that condition.

In ballet dancers, forcing turnout or ‘fishing’ the foot (Fig. 40.11) can predispose to this condition. Forced turnout occurs when the feet are planted in greater external rotation than is available at the hips—this places excessive rotary forces at the posterior ankle (and knee). ‘Fishing’ is a ballet term referring to foot deviation away from the ideal leg/foot alignment as shown in Figure 40.11.

Technique assessment, and strength and endurance training of the hip external rotators, calf complex and foot intrinsic muscles are an essential part of treatment. Treatment of posterior impingement syndrome includes modified workload, manual mobilisation of the subtalar, talocrural and midfoot joints as well as NSAIDs or paracetamol (acetaminophen) for symptomatic relief. Ballet dancers can cross-train via Pilates and gradually



(a)



(b)

Figure 40.10 Anatomy that underpins posterior impingement when symptoms are present. These anatomical changes can be asymptomatic (a) A prominent posterior process of the talus (arrow) (b) The os trigonum (arrow)

progress their training from working flat (no raising) to *en pointe* during the treatment phase.

If the condition persists, a corticosteroid injection around the area of maximal tenderness may reduce pain. This is best done from the lateral side, as the medial aspect of the ankle contains the neurovascular bundle.

If posterior impingement does not respond to conservative management, then arthroscopic or open removal of the enlarged posterior process or the os trigonum is indicated. For dancers, the modal time it takes to return to full performance is around 4 months.

SEVER'S DISEASE

Sever's disease or calcaneal apophysitis is a common insertional enthesopathy among adolescents (Chapter 44). It can be considered the Achilles tendon equivalent of Osgood-Schlatter disease at the patellar tendon insertion. It should be treated by reducing load until pain is settled, (a heel raise is helpful) increasing strength and endurance, and then gradually reintroducing load.

OTHER CAUSES OF PAIN IN THE ACHILLES REGION (GRADUAL ONSET)

Accessory soleus

A rare cause of Achilles region pain in the sports medicine setting, an accessory soleus is present in about 5-10% of people.^{131, 132} The condition is more common among men than women, and average age of presentation is 20 years. The true accessory soleus inserts into the calcaneum, but it may also insert into the distal part of the tendon. It generally occurs unilaterally, but it may occur bilaterally.¹³³

The primary presenting patterns are pain in the Achilles region during exercise (a 'compartment' type pain) with swelling, or a painless swelling. There is an obvious muscle bulk very low, deep to the Achilles, on both the medial and lateral sides which is often tender on palpation. The main differential diagnosis is soft tissue tumour. Imaging findings are characteristic; plain radiographs have poor sensitivity for this condition but, when positive, show a soft tissue shadow posterior to the tibia obscuring the pre-Achilles fat pad. MRI is the investigation of choice¹³² and confirms a mass with the same signal intensity as muscle; ultrasound can also confirm that there is a low-lying soleus (or accessory) soleus. In cases that are a symptomatic, observation is an appropriate treatment, but if symptoms warrant, surgical removal of the accessory soleus is probably the best treatment.¹³⁴ Botulinum toxin has been trialled in a case series.¹²³

Referred pain

Referred pain to this region from the lumbar spine or associated neural structures is unusual, but always warrants consideration in challenging cases.



(a)



(b)

Figure 40.11 'Fishing' at the ankle can predispose to posterior impingement in a ballet dancer (a) 'Fished' position. The midline (second metatarsal) of the foot deviates laterally to the midline of the ankle (b) Correct demi-pointe position. The midline of the foot is aligned with the midline of the ankle

ACUTE ACHILLES TENDON RUPTURE (COMPLETE)

Clinical approach

Complete rupture of the Achilles tendon (Fig. 40.12) classically occurs in athletes in their 30s or 40s, and the male:female ratio is 10:1. The typical patient is a 40-year-old sports-active male. The incidence is reported to be 5–36 in 100 000 and is increasing, especially over the last 10 years, with 60–90% of all Achilles tendon ruptures occurring during sports. The majority of the patients have not had any symptoms from the Achilles tendon region prior to injury. We address the debate around conservative versus surgical treatment of Achilles rupture below. Regardless of treatment, long-term deficits in strength and function are common and approximately only 30–40% of athletes return to their previous pre-injury level.¹³⁵

History

Usually the rupture occurs when the person performs a quick change of direction and the ankle is forced into dorsiflexion while the calf muscle contracts. The patient describes feeling 'as if I was hit or kicked in the back of the leg'; pain might not always be the strongest sensation. This is immediately followed by grossly diminished function. A snap or tear may be audible and the patient has difficulty walking. However, the ability to resume walking with short steps returns quickly, but without power in the push-off phase.

The patient will usually have an obvious limp, but may have surprisingly good function through the use of compensatory muscles such as plantaris and the long toe flexors. That is, the patient may be able to walk, but not on



Figure 40.12 Achilles tendon rupture presents very differently from overuse Achilles tendinopathy. This intraoperative photograph highlights the severity of this injury. Not all cases need to be operated on (see below)

his or her toes. During the initial assessment, the clinician should take time to ask about risk factors for Achilles tendon rupture (including previous Achilles rupture, other major tendon rupture, recent local corticosteroid injection, or clues to other collagen diseases, diabetes mellitus type 2, etc.).¹³⁶

Physical examination

Four clinical tests greatly simplify examination for complete Achilles tendon rupture (Fig. 40.3):

1. The calf squeeze test (also known as Simmond's or Thompson's test) (see Fig. 40.3h) has a sensitivity of 0.96, a specificity of 0.93, a positive likelihood ratio of 13.7 and a negative likelihood ratio of 0.04.¹⁰
2. Matle's test has a sensitivity of 0.88, a specificity of 0.85, a positive likelihood ratio of 6.3 and a negative likelihood ratio of 0.14.¹⁰
3. Copeland's test has a sensitivity of 0.78 (specificity and likelihood ratios are unreported).¹⁰
4. Palpation for a gap in the tendon has a sensitivity of 0.73, a specificity of 0.89, a positive likelihood ratio of 6.6 and a negative likelihood ratio of 0.3.¹⁰ The gap in the tendons becomes increasingly difficult to palpate as time between rupture and clinical assessment increases.

Key outcome measures

The Achilles tendon total rupture score (ATRS) is a patient-reported outcome measure developed to evaluate outcome after an Achilles tendon rupture.^{137, 138} ATRS evaluates aspects of symptoms and physical activity. The score consists of 10 questions, where the score for each item ranges from 0–10 on a Likert scale, with a maximum score of 100. It has been shown to have excellent reliability (ICC 0.98–0.99), an effect size of 0.87–2.21 and a minimal detectable change of 6.75 points. The score in patients varies from 17–100 depending on the stage of

recovery (approximately a mean of 40/100 at 3 months, 70/100 at 6 months and 80–100/100 at 12 months).

A heel raise test to measure both calf-muscle endurance and the maximal heel raise height is commonly used to measure outcome in this population. For this test the person is asked to perform as many high single-leg heel raises at a standardised frequency (usually 30 reps/min) as possible to evaluate the endurance of the calf muscles. The maximal heel raise height is measured by evaluating how high the patient can lift the heel off the floor. This test has good reliability (ICC 0.78–0.84) and has been used to evaluate recovery over time in patients with Achilles tendon rupture.^{137, 139–143} The side-to-side differences in maximal heel raise height, when comparing the injured and healthy sides in patients with an Achilles tendon rupture, correlated¹⁴¹ with the occurrence of tendon elongation on the injured side.

Outcome measures that evaluate higher-level function, such as single-leg jump tests for distance or height, are used later in rehabilitation, and the performance on the injured side can be compared to the healthy side.

Investigations

Investigations such as ultrasonography and magnetic resonance imaging (MRI) are sometimes used as adjuncts to clinical investigation. However, such investigations are seldom necessary if the clinical history and clinical examination are thorough. Ultrasonography and MRI are needed only in doubtful cases, such as re-ruptures and when there is a long delay from the index injury. However, they are not needed in routine healthcare to establish the diagnosis of a torn Achilles tendon.

Rehabilitation of Achilles tendon ruptures

Historically, there was a fear of loading the Achilles tendon during the initial 6–8 weeks because of a putative risk of re-rupture and other complications. Systematic

Treatment of the acutely ruptured Achilles tendon: To operate or not to operate

Surgical management

With a 2–3% re-rupture rate, open surgical treatment of Achilles tendon rupture has approximately 10% lower risk of re-rupture compared with non-surgical treatment, which stands at just under 13%.¹⁴⁴ Promising new surgical techniques may further reduce the rate of postoperative re-rupture (see below). However, open operative treatment is associated with an increased risk of complications, including superficial wound infection, adhesions and disturbed skin sensitivity.¹⁴⁵ Another approach to reduce these surgical

complications is to perform surgery percutaneously, but this does not entirely eliminate the risk of complications.

The postoperative treatment regimen has a significant effect on the outcome.¹⁴⁶ Comparisons of rigid cast immobilisation with a short period of rigid cast and then use of a functional brace after surgical intervention, indicates that early mobilisation decreases re-rupture rates without an increased risk of other complications. Furthermore, patient satisfaction is higher with the functional brace. Postoperative management is usually with an immobilising brace for the

first 6–8 weeks; however, the period of postoperative immobilisation may very well be shortened and recent studies have used short or even no immobilisation after surgery. Because range of movement and strength can be difficult to regain after rupture repair, the earliest possible mobilisation and rehabilitation is recommended.

Surgery using stable core sutures and local augmentation has good results with no re-ruptures.¹⁴⁷ This technique is based on the principle of using the strongest possible suture material to make the surgical construct as strong as possible and enable the patient to start early range-of-motion training, even without any immobilisation at all. Early weight-bearing is encouraged as well.

Non-surgical management

Historically, non-surgical management of an Achilles tendon rupture has been recommended in older patients or patients with low levels of activity.¹⁴⁴ Early mobilisation for non-surgically treated patients also decreases the re-rupture rate compared with rigid cast immobilisation.¹⁴⁶ When both surgically and non-surgically treated patients received identical mobilisation protocols (cast for 2 weeks and then a functional brace), there was no difference in patient-reported outcome. Non-surgical treatment leads to a high success rate, provided no re-rupture occurs, and may therefore be considered a valid option for all patients (Fig. 40.13).



Figure 40.13 There is increasing evidence that non-surgical management with appropriate rehabilitation can be effective

B

reviews¹⁴⁶ report that simple and early mobilisation while in the functional brace improved outcomes. Thus, patients are often instructed to move the toes and perform light plantar flexion isometrics or active range of motion (if the brace allows for it). This advice applies whether the patient has undergone surgical repair of the tendon or is being managed non-surgically.

After the removal of the brace or cast, the goal for the patient is to regain calf-muscle strength and achieve normal gait while limiting the chance of re-rupture. It is usually at this stage that the formal rehabilitation starts. It is important to understand that the load on the Achilles tendon can be increased both by increasing the external load and by increasing the speed of movement. Therefore, during the initial 4 weeks of rehabilitation, slower and controlled movements are recommended. All the muscles around the ankle need to be strengthened and this can be progressed through the use of exercise tubing (Fig. 40.14a). Moreover, the calf muscle is strengthened by progressively increasing the load through sitting heel raises (Fig. 40.14b) and bilateral heel raises, and ultimately the patient should be able to perform a single-leg heel raise.

PRACTICE PEARL

Achieving full ankle range of motion (as compared with the healthy side) is important in all cases of Achilles rupture treatment. It should not be achieved by stretching the calf musculature in the first 3–4 weeks in the rehabilitation period. Such stretching can leave the Achilles tendon to heal in an elongated position.^{147, 148} Instead, full ankle range of motion can be achieved through joint mobilisation and stretching into dorsiflexion with the knee in a flexed position.

Timing to return to jogging and sports

Whether or not treatment is surgical or non-surgical, the decision to return to jogging and sports is often time-based. Traditional recommendations are that jogging can be started after 12–16 weeks, return to non-contact sports after 16–20 weeks and contact sports after 20–24 weeks.

Time criteria may be inadequate, and it is essential that the key criteria for return to jogging and sports be functional, such as the recovery of the capacity of the tendon to store and release energy in sports-specific



(a)



(b)

Figure 40.14 Calf muscle strengthening after Achilles tendon rupture (a) Early rehabilitation includes low-load ankle plantarflexion with knee extended (e.g. with Thera Band™) (b) Seated heel raise against resistance loads the soleus preferentially

loading, as well as the patient having excellent calf-muscle strength and ankle range of motion. Muscle bulk may lag many months behind these functional criteria and should not be used to delay return to sport. The patient should be able to hop similarly to the unaffected side, and repeatedly store and release energy (repeated hopping) in the tendon adequately for their sport. Strength, measured with single-leg heel raises, should be at least 20 reps but can range from 6–70 reps in healthy individuals.¹⁴⁹

Longer-term rehabilitation deficits

Regardless of whether initial treatment was surgical or non-surgical, complications such as calf-muscle weakness, tendon elongation and gait abnormalities can persist for at least a year after injury. Because of individual differences, rehabilitation should be tailored to each patient's deficits. Often the patient cannot raise the heel as high on the injured side as the uninjured side. The clinician should test whether this is due to tendon elongation, weakness or both. This does not appear to limit return to physical activity in the majority of patients, and whether it has any long-term implications is not known.

About 50% of patients can perform a single-leg heel raise 3 months after their injury. Between 3 and 6 months after injury, the majority of patients can achieve this milestone. However, it is common to have a strength deficit of 10–30% in the calf musculature on the injured side compared with the healthy side, and this commonly becomes permanent.

Thus, it is important to encourage the patient to persist with a comprehensive functional rehabilitation program and ideally return to full activity. The clinician needs to be alert to identifying any specific barriers to this successful return to activity. The early phase of rehabilitation in particular should be closely monitored. Studies have shown very limited improvement between 1 and 2-year follow-ups.

Additional studies are needed to garner better evidence about the appropriate and most beneficial type of exercises both in the initial immobilisation period (if any) and during rehabilitation. The issue of tendon lengthening after treatment, especially non-surgical treatment, leading to insufficient power in the calf muscles is still not well studied.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e